

How to write the proposal:

1. Reading 20 research paper
 - a. Semantic Scholar
 - b. AI

To find research papers

2. Purpose of LR at this point is to identify the research gap
3. Summarize them in tabular form

[Create the table in spreadsheet](#)

4. Where will you start when writing the proposal?
 - a. **Research gap**

Title for the research gap: describe in paragraph format using citation because research gap comes from literature – research paper

This description must justify why do you call this a research gap with citation from literature

Note: Check the following email for template

If you are planning to address more than one gap then repeat for each gap the above.

----- Forwarded message -----

From: **Guhanathan Poravi** <guhanathan.p@iit.ac.lk>

Date: Fri, 23 Jan 2026 at 19:05

Subject:    MSc ASE & MSc CSF
Final Project Jan 2026 Starter – Supervisor Selection, Topic Finalisation

and Official Milestone Schedule
To:   

- b. Contribution to Body of Knowledge

By addressing those research gap (coming up with solution) what contribution am I going to make to the Body of Knowledge

Heart Attack Prediction System – that is what my research is about then

Problem domain is **Cardiology**

What new thing you will addressing that was overlooked before.
Example in Sri Lanka no one looked at Sri Lankan Specific Attributes and you are first person – that will Contribution to problem Domain

Technology domain is **ML / DL / NN / ???**

New architecture if it is CNN, New ensemble if it is ML, DL, ??, Multimodal, XAI – with contribution to architecture

- i. Contribution to Problem Domain

Topic: Justification why is a contribution

- ii. Contribution to Technology Domain (Research Domain)

Topic: Justification why is a contribution

No research gap it is not a research project – FAIL

Identified research gap but no contribution to BoK – maximum 50s

There is a significant contribution to Problem Domain – maximum 69

To cross above 69 you must have contribution to Technology Domain

If its cyber security project – Problem Domain and Technology Domain

Problem Domain is Malware

Technology Domain is AI

- c. Research challenge

This is not about the personal challenge but research challenge – what is it? Anyone undertaking this research what kind of challenges they will face to complete

Topic for the Challenge: why do you say it is a challenge

NOT PERSONAL CHALLENGE – I don't know ML, I don't know Cyber Security, I have work, family and kids those are personal challenge and we are not talking about them

d. Research question

Only one research question – answer that question is the research you are carrying out and your thesis + prototype is the answer to the question

Can have sub questions

Main Question

RQ:

Sub Question

Sub RQ1:

Sub RQ2:

Sub RQ3:

Maximum of three sub question and sub question must be the breakdown of the main questions

<https://www.scribbr.com/research-process/research-questions/>

e. Research Aim - must be one sentence

The aim of the research project is to design, develop and evaluate a model to predict heart attack by taking Sri Lankan specific attributes.

Then in the next paragraph you can elaborate the aim in detail.

f. Research Objectives

Check list – all possible things that I need to do in order to complete this project

1. To identify the research gap in the area of problem domain
2. To conduct the Literature review for
3. To design the architecture
4. To implement
5. To evaluate
6. To write and submit the thesis ...
7. To publish

g. Then LR

Chapter 02: Literature Review

Cannot give a complete set of headings and subheading like I can do for the other chapters because LR is specific to topic

But broader list of topics can be given

2.1 Chapter Overview

2.2 Concept Map

2.3 Problem Domain

Last sub topic under the problem domain must be the architecture of the proposed solution – if it is ML then pipeline

2.4 Existing Work

2.5 Technology review – here each part of ML pipeline must be discussed in detail – what are the choices – what are the alternatives – pros and cons – what and all could be the possible choice for your project

2.6 Evaluation and Benchmarking

2.7 Chapter Summary